

Supplement to “The Confidence Cost of Pairwise Distribution Learning”

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This supplement gives complete proofs of the winner-chain occupation bound, the arbitrary-start upper bound, and the adaptive confidence lower bound. It also proves the bounded-menu lower bound stated as an open-direction result in the main paper and records the numerical and source audits. Numbering is local to the supplement.

1 Pairwise Winner Chain

Fix a distribution $p \in \Delta_n$. From state i , the chain chooses $j \neq i$ uniformly, queries the pair $\{i, j\}$, and makes the returned label the next state. If $p_i + p_j > 0$, then

$$P(i, j) = \frac{p_j}{(n-1)(p_i + p_j)}, \quad i \neq j. \quad (1)$$

If $p_i = p_j = 0$, the pair response may follow any fixed convention. The diagonal is the remaining row mass.

Lemma 1 (Stationarity, reversibility, and minorization). *The distribution p is stationary. On its positive support the chain is reversible, and for every state i ,*

$$P(i, \cdot) \geq \frac{1}{n-1} p(\cdot) \quad (2)$$

coordinatewise.

Proof. For distinct positive-mass states i, j ,

$$p_i P(i, j) = \frac{p_i p_j}{(n-1)(p_i + p_j)} = p_j P(j, i).$$

Transitions from a positive-mass state to a zero-mass state have probability zero. Detailed balance on the support proves reversibility and stationarity.

For the minorization, first take $j \neq i$. If $p_i + p_j > 0$, then

$$P(i, j) = \frac{1}{n-1} \frac{p_j}{p_i + p_j} \geq \frac{p_j}{n-1},$$

since $p_i + p_j \leq 1$. If both masses vanish, the right side is zero. If $p_i = 0$, the diagonal inequality is immediate. If $p_i > 0$, then against every challenger j , the probability of remaining at i is at least $p_i/(p_i + p_j) \geq p_i$. Averaging over challengers gives $P(i, i) \geq p_i \geq p_i/(n-1)$. $\square$

For $n > 2$, the minorization gives a Markov kernel R such that

$$P = \frac{1}{n-1} \Pi_p + \left(1 - \frac{1}{n-1}\right) R, \quad (3)$$

where every row of Π_p is p . Thus, for any initial law μ ,

$$d_{\text{TV}}(\mu P^B, p) \leq \left(1 - \frac{1}{n-1}\right)^B \leq e^{-B/(n-1)}. \quad (4)$$

For $n = 2$, one query returns an exact p draw from either state.

2 Positive Spectrum and Coordinate Variance

We work on the positive support of p . Zero-mass coordinates contribute nothing to TV under stationarity. Let $m = |\text{supp}(p)|$. For a function $f \in L_2(p)$, reversibility gives the Dirichlet form

$$\langle f, (I - P)f \rangle_p = \sum_{1 \leq i < j \leq m} \frac{p_i p_j}{(n-1)(p_i + p_j)} (f_i - f_j)^2. \quad (5)$$

Lemma 2 (Positive semidefinite kernel). *Every eigenvalue of P on $L_2(p)$ lies in $[0, 1]$.*

Proof. The Markov and reversibility properties put the spectrum in $[-1, 1]$. For each positive pair,

$$\frac{p_i p_j}{p_i + p_j} (f_i - f_j)^2 \leq p_i f_i^2 + p_j f_j^2. \quad (6)$$

Indeed, the right side minus the left side is $(p_i f_i + p_j f_j)^2 / (p_i + p_j)$. Each coordinate appears in at most $m - 1$ positive pairs. Summing (6) and dividing by $n - 1$ gives

$$\langle f, (I - P)f \rangle_p \leq \langle f, f \rangle_p.$$

Hence $\langle f, Pf \rangle_p \geq 0$ for every f . □

For coordinate i , put $f_i(x) = \mathbf{1}\{x = i\} - p_i$ and

$$H_i = \langle f_i, (I - P)^{-1} f_i \rangle_p, \quad (7)$$

where the inverse acts on the mean-zero subspace.

Lemma 3 (Finite-horizon coordinate variance). *For a stationary length- T trajectory,*

$$\text{Var}_p \left(\frac{1}{T} \sum_{t=1}^T f_i(X_t) \right) \leq \frac{2H_i}{T}. \quad (8)$$

Proof. Expand f_i in an orthonormal eigenbasis of the mean-zero subspace. If a^2 is the squared coefficient associated with eigenvalue $\rho \in [0, 1)$, its contribution to the variance is

$$\frac{a^2}{T^2} \left(T + 2 \sum_{s=1}^{T-1} (T-s) \rho^s \right) \leq \frac{2a^2}{T(1-\rho)}.$$

Summing the contributions gives (8) by the spectral definition of H_i . □

3 An Explicit Comparison Green Function

Relabel the positive probabilities so that $p_1 \geq \dots \geq p_m > 0$. Define the graph Laplacian L_0 on $[m]$ whose edge conductances are

$$w_{ij}^0 = \frac{\min\{p_i, p_j\}}{n}. \quad (9)$$

For $k \in [m]$, define

$$c_k = kp_k + \sum_{j>k} p_j = \sum_{j=1}^m \min\{p_j, p_k\}. \quad (10)$$

Lemma 4 (Helmert diagonalization). *For $k = 2, \dots, m$, let*

$$u_k = \frac{\underbrace{(1, \dots, 1)}_{k-1}, -(k-1), 0, \dots, 0}{\sqrt{k(k-1)}}. \quad (11)$$

The vectors $(u_k)_{k=2}^m$ are an orthonormal basis of the Euclidean zero-sum subspace and

$$L_0 u_k = \frac{c_k}{n} u_k. \quad (12)$$

Proof. Orthonormality is the standard Helmert calculation. Fix k . Every coordinate larger than k sees equal conductance p_j/n to all first k coordinates, whose entries in u_k sum to zero. Its image is therefore zero. A coordinate $i < k$ differs only from coordinate k among the first k , contributing kp_k/n . Every coordinate $j > k$ contributes p_j/n . Thus the i th image is $(c_k/n)u_k(i)$. The same sum at coordinate k is multiplied by $-(k-1)$, which proves (12). $\square$

Let $D = \text{diag}(p)$. The graph Laplacian associated with (5) is $L = D(I - P)$. For the centered indicator f_i ,

$$Df_i = p_i(e_i - p) =: b_i. \quad (13)$$

Therefore

$$H_i = b_i^\top L^\dagger b_i. \quad (14)$$

The exact conductance of L satisfies

$$\frac{p_i p_j}{(n-1)(p_i + p_j)} \geq \frac{\min\{p_i, p_j\}}{2(n-1)} \geq \frac{1}{2} w_{ij}^0. \quad (15)$$

It follows by the variational characterization of inverse Laplacians that

$$H_i \leq 2b_i^\top L_0^\dagger b_i. \quad (16)$$

Lemma 5 (Exact diagonal formula). *Define*

$$\begin{aligned} A_i &= \sum_{k=2}^{i-1} \frac{(1 - c_k)^2}{k(k-1)c_k} + \mathbf{1}\{i \geq 2\} \frac{(i - c_i)^2}{i(i-1)c_i} \\ &\quad + \sum_{k=i+1}^m \frac{c_k}{k(k-1)}. \end{aligned} \quad (17)$$

Then

$$b_i^\top L_0^\dagger b_i = np_i^2 A_i, \quad H_i \leq 2np_i^2 A_i. \quad (18)$$

Proof. First observe from (10) and (11) that

$$u_k^\top p = \frac{1 - c_k}{\sqrt{k(k-1)}}. \quad (19)$$

Since $b_i = p_i(e_i - p)$, its projection onto u_k is p_i times

$$u_k(i) - u_k^\top p = \begin{cases} -(1 - c_k)/\sqrt{k(k-1)}, & k < i, \\ -(i - c_i)/\sqrt{i(i-1)}, & k = i, \\ c_k/\sqrt{k(k-1)}, & k > i. \end{cases} \quad (20)$$

Divide the squared projections by the eigenvalues c_k/n from Lemma 4 and sum over k . This gives the equality in (18). The inequality follows from (16). $\square$

4 The Profile Sum Telescopes

Write $A_i = L_i + M_i + R_i$ for the three terms in (17). We prove the aggregate inequality used in the main paper.

Lemma 6 (Uniform profile bound). *For every nonincreasing positive probability vector,*

$$\sum_{i=1}^m p_i \sqrt{A_i} \leq 2 + 2\sqrt{2}. \quad (21)$$

Proof. By Jensen's inequality and a change in summation order,

$$\sum_i p_i \sqrt{L_i} \leq \sqrt{\sum_i p_i L_i} \quad (22)$$

$$= \sqrt{\sum_{k=2}^{m-1} \frac{(1-c_k)^2}{k(k-1)c_k} \sum_{i>k} p_i} \quad (23)$$

$$\leq \sqrt{\sum_{k=2}^{m-1} \frac{1}{k(k-1)}} \leq 1. \quad (24)$$

The last line uses $\sum_{i>k} p_i \leq c_k$ and $1 - c_k \leq 1$. Similarly,

$$\sum_i p_i \sqrt{R_i} \leq \sqrt{\sum_i p_i R_i} \quad (25)$$

$$= \sqrt{\sum_{k=2}^m \frac{c_k}{k(k-1)} \sum_{i<k} p_i} \quad (26)$$

$$\leq \sqrt{\sum_{k=2}^m \frac{1}{k(k-1)}} \leq 1. \quad (27)$$

For the middle term, define $S_i = \sum_{j \geq i} p_j$. Since $c_i = ip_i + \sum_{j>i} p_j \geq S_i$ and $i/(i-1) \leq 2$,

$$\sum_i p_i \sqrt{M_i} \leq \sqrt{2} \sum_i \frac{p_i}{\sqrt{c_i}} \leq \sqrt{2} \sum_i \frac{S_i - S_{i+1}}{\sqrt{S_i}}. \quad (28)$$

For $0 \leq b \leq a$,

$$\frac{a-b}{\sqrt{a}} \leq 2(\sqrt{a} - \sqrt{b}). \quad (29)$$

Apply this with $a = S_i, b = S_{i+1}$ and telescope to obtain

$$\sum_i p_i \sqrt{M_i} \leq 2\sqrt{2}. \quad (30)$$

Finally, $\sqrt{L_i + M_i + R_i} \leq \sqrt{L_i} + \sqrt{M_i} + \sqrt{R_i}$. Combining (24), (27), and (30) proves the claim. $\square$

5 Stationary Occupation Risk

Theorem 7 (Stationary risk). *For a stationary length- T winner-chain trajectory with empirical law $\hat{p}_T$,*

$$\mathbb{E} d_{\text{TV}}(\hat{p}_T, p) \leq (2 + 2\sqrt{2}) \sqrt{\frac{n}{T}}. \quad (31)$$

Proof. If the support has size one, the claim is immediate. Otherwise, Lemmas 3 and 5 give

$$\text{Var}(\hat{p}_T(i)) \leq \frac{4np_i^2 A_i}{T}.$$

Jensen's inequality and Lemma 6 yield

$$\begin{aligned} \mathbb{E} d_{\text{TV}}(\hat{p}_T, p) &\leq \frac{1}{2} \sum_i \mathbb{E} |\hat{p}_T(i) - p_i| \\ &\leq \frac{1}{2} \sum_i \sqrt{\text{Var}(\hat{p}_T(i))} \\ &\leq \sqrt{\frac{n}{T}} \sum_i p_i \sqrt{A_i} \leq (2 + 2\sqrt{2}) \sqrt{\frac{n}{T}}. \end{aligned}$$

Zero-mass coordinates are never visited under stationarity and add zero. $\square$

6 Arbitrary Starts and High Confidence

Set $C_0 = 2 + 2\sqrt{2}$ and

$$B = \left\lceil (n-1) \log \frac{48}{\varepsilon} \right\rceil, \quad T = \left\lceil 48^2 C_0^2 \frac{n}{\varepsilon^2} \right\rceil. \quad (32)$$

Start the winner chain from an arbitrary state and form its empirical law $\hat{p}$ over the T states after burn-in.

Lemma 8 (One base estimate). *The estimate satisfies*

$$\mathbb{P}\{d_{\text{TV}}(\hat{p}, p) > \varepsilon/4\} \leq 1/6. \quad (33)$$

Proof. By (4), the state after burn-in is within $\varepsilon/48$ TV of stationarity. Generate the entire subsequent path by applying one common path kernel to either initial law. Data processing shows that the arbitrary-start and stationary path laws remain within $\varepsilon/48$ TV. Since TV loss lies in $[0, 1]$, their expected losses differ by at most $\varepsilon/48$. Theorem 7 and the choice of T give

$$\mathbb{E}d_{\text{TV}}(\hat{p}, p) \leq \varepsilon/48 + \varepsilon/48 = \varepsilon/24.$$

Markov's inequality proves (33). $\square$

Repeat this construction independently R times and let $\hat{p}_r$ be the candidates. Define an L_1 geometric median

$$\tilde{p} \in \arg \min_{q \in \Delta_n} \sum_{r=1}^R d_{\text{TV}}(q, \hat{p}_r). \quad (34)$$

Existence follows from compactness. The optimization is a linear program after introducing one absolute-value slack variable per candidate and coordinate.

Lemma 9 (Metric median). *In any metric space, suppose $G > R/2$ of the points $z_1, \dots, z_R$ lie within distance r of p . Every minimizer z of the sum of distances to the points satisfies*

$$d(z, p) \leq \frac{2G}{2G - R} r. \quad (35)$$

Proof. Let $\mathcal{G}$ denote the good indices and $d = d(z, p)$. For a good point,

$$d(z, z_r) - d(p, z_r) \geq d - 2r.$$

For a bad point, the triangle inequality gives $d(z, z_r) - d(p, z_r) \geq -d$. Since the sum at z is no larger than the sum at p ,

$$0 \geq G(d - 2r) - (R - G)d = (2G - R)d - 2Gr.$$

Rearranging proves the result. $\square$

Theorem 10 (High-confidence upper bound). *If $R = \lceil 18 \log(1/\delta) \rceil$, then*

$$\mathbb{P}\{d_{\text{TV}}(\tilde{p}, p) > \varepsilon\} \leq \delta. \quad (36)$$

The total number of pair queries is $O(n \log(1/\delta)/\varepsilon^2)$.

Proof. Every candidate is good with probability at least $5/6$ by Lemma 8. Hoeffding's inequality gives

$$\mathbb{P}\{G < 2R/3\} \leq e^{-R/18} \leq \delta.$$

On the complementary event, apply Lemma 9 with $r = \varepsilon/4$. The multiplicative factor in (35) is at most four, so the output error is at most ε .

Each repetition uses $B+T$ queries. For $0 < \varepsilon \leq 1/16$, $\log(48/\varepsilon) = O(1/\varepsilon^2)$, hence $B+T = O(n/\varepsilon^2)$. Multiplying by R proves the query bound. $\square$

7 Adaptive Confidence Lower Bound

Let $n \geq 4$, $0 < \varepsilon \leq 1/16$, and define

$$p_a(1) = a, \quad p_a(j) = \frac{1-a}{n-1}, \quad j \geq 2. \quad (37)$$

Take $a_- = 1/2 - 2\varepsilon$ and $a_+ = 1/2 + 2\varepsilon$. Both lie in $[1/4, 3/4]$, and

$$d_{\text{TV}}(p_{a_-}, p_{a_+}) = |a_- - a_+| = 4\varepsilon. \quad (38)$$

If a queried pair excludes label 1, its response is uniform under both hypotheses. If the pair is $\{1, j\}$, write one for a light-label response. The Bernoulli parameter is

$$r(a) = \frac{1-a}{1+(n-2)a}. \quad (39)$$

For $a \in [1/4, 3/4]$ and $n \geq 4$,

$$\frac{1}{3n} \leq r(a) \leq \frac{3}{n}, \quad |r'(a)| = \frac{n-1}{(1+(n-2)a)^2} \leq \frac{16}{n}. \quad (40)$$

Also $1 - r(a) \geq 1/4$. By the mean value theorem and the Bernoulli inequality

$$D_{\text{KL}}(\text{Bern}(x) \parallel \text{Bern}(y)) \leq \frac{(x-y)^2}{y(1-y)}, \quad (41)$$

every possible pair query has conditional KL divergence at most

$$C_1 \frac{\varepsilon^2}{n} \quad (42)$$

between the two response laws, for a universal C_1 .

Theorem 11 (Adaptive lower bound). *Any possibly randomized and adaptive learner using a fixed budget Q and satisfying*

$$\sup_{p \in \Delta_n} \mathbb{P}_p\{d_{\text{TV}}(\hat{p}, p) > \varepsilon\} \leq \delta \quad (43)$$

must have

$$Q \geq c \frac{n \log(1/\delta)}{\varepsilon^2} \quad (44)$$

for $0 < \delta \leq 1/8$ and a universal $c > 0$.

Proof. Include the learner's private random seed in its transcript. Conditional on every past transcript, its next pair is fixed. The conditional response laws then obey (42), regardless of which pair was selected. The KL chain rule gives

$$D_{\text{KL}}(\mathbf{P}_-^Q \parallel \mathbf{P}_+^Q) \leq C_1 Q \frac{\varepsilon^2}{n}, \quad (45)$$

where $\mathbf{P}_-^Q, \mathbf{P}_+^Q$ are the complete transcript laws.

Given the output $\hat{p}$, test the two hypotheses by choosing the closer of p_{a_-} and p_{a_+} in TV. By (38), every estimate within ε of the truth is strictly closer to that truth. Thus the two testing errors are at most δ . The Bretagnolle–Huber inequality (Tsybakov, 2009) gives

$$2\delta \geq \frac{1}{2} \exp\{-D_{\text{KL}}(\mathbf{P}_-^Q \parallel \mathbf{P}_+^Q)\}. \quad (46)$$

Combining (45) and (46) yields

$$Q \geq \frac{n}{C_1 \varepsilon^2} \log \frac{1}{4\delta}.$$

For $\delta \leq 1/8$, this is (44) after changing the universal constant. $\square$

Theorem 10 and Theorem 11 together prove the minimax theorem in the main paper.

8 A Bounded-Menu Confidence Lower Bound

The same family quantifies how menu size can reduce the confidence obstruction. Suppose every allowed conditional query has cardinality at most k , where $2 \leq k \leq n$.

Proposition 12 (Bounded-menu lower bound). *Under the parameter ranges of Theorem 11, every learner from menus of size at most k requires*

$$Q = \Omega\left(\frac{n \log(1/\delta)}{k \varepsilon^2}\right). \quad (47)$$

Proof. Use the two distributions in (37). A menu excluding label 1 is uniform under both hypotheses and carries no information. If a menu contains label 1 and $s \leq k - 1$ light labels, the identity of a light winner is uniform conditional on a light outcome. The only informative statistic is therefore Bernoulli with light-outcome probability

$$r_s(a) = \frac{s(1-a)}{(n-1)a + s(1-a)}. \quad (48)$$

Uniformly for $a \in [1/4, 3/4]$, $r_s(a) = \Theta(s/n)$ and

$$|r'_s(a)| = \frac{s(n-1)}{((n-1)a + s(1-a))^2} = O(s/n). \quad (49)$$

Equation (41) now bounds the one-query KL by $Ck\varepsilon^2/n$. The adaptive transcript and testing argument from Theorem 11 gives (47). $\square$

Ordinary distribution-learning hardness also gives $\Omega(n/\varepsilon^2)$ even when the full domain is queryable (Canonne, 2020). Hence every bounded-menu learner has lower bound

$$\Omega\left(\frac{n + (n/k) \log(1/\delta)}{\varepsilon^2}\right).$$

We do not claim a matching upper bound when $k > 2$.

9 Mechanical Verification

The proofs are analytic. The script `verify_trajectory_theorem.py` performs independent floating-point checks designed to catch indexing, scaling, and inequality-direction errors. It does not supply evidence needed by any theorem.

Across 1,014 seeded mass profiles in dimensions 2 through 40, including uniform, one-heavy, geometric, and Dirichlet profiles, the script checked:

1. the winner kernel’s symmetrized eigenvalues lie in $[0, 1]$,
2. the Helmert formula (18) agrees with a direct pseudoinverse of L_0 ,
3. the exact Green function is bounded by twice the comparison Green function coordinatewise,
4. each of the left, middle, and right profile sums obeys its analytic bound,
5. the exact finite- T stationary covariance for $T \in \{1, 2, 5, 20, 100\}$ obeys Lemma 3, and
6. the normalized one-query KL in the hard family stays below the loose analytic constant over $4 \leq n \leq 256$ and 80 logarithmically spaced accuracies.

The largest observed normalized KL, $nD_{\text{KL}}/\varepsilon^2$, was 161.847400. The terminal output was `all trajectory-theorem checks passed`.

An additional stress script sampled highly imbalanced simplex points through dimension 1024. The observed asymptotic L_1 risk divided by $\sqrt{n/T}$ remained below 2.28, while the theorem uses the conservative TV constant $2 + 2\sqrt{2}$. These searches motivated no step of the proof.

10 Source and Novelty Audit

We read the full 34-page paper of Kleinberg et al. (2026). Its generic complete-family learner runs a global winner chain, burns in, keeps one terminal state, and independently restarts. Its hierarchical learner keeps local trajectories but estimates one binary tree split at a time. Its pairwise corollary is $O(n \log^2 n \log(n/\delta)/\varepsilon^2)$, and its conclusion explicitly leaves the polylogarithmic gaps open. It contains neither the empirical-law risk theorem nor the multiplicative-confidence lower bound.

We also read the full papers on accelerated spectral ranking (Agarwal et al., 2018), graph-resistance estimation (Hendrickx et al., 2019), and the asymptotic BTL minimax rate (Hendrickx et al., 2020), together with the full paper and supplement of Shah et al. (2015). They study fixed comparison designs and parameter losses. Their useful uniform guarantees assume positive weights and bounded parameter ranges or enter a many-comparisons-per-edge regime. None analyzes the query-driven winner trajectory used here.

The full exact-sampling paper of Fotakis et al. (2022) uses coupling from the past for an exogenous pair design. Its distribution-free procedure first estimates all weights to relative accuracy under a learnability condition, then modifies the chain and performs exact correction. Its goal and complexity are distinct from minimax labeled TV learning.

For statistical context, we read the complete ordinary distribution-learning note of Canonne (2020), the full Markov Bernstein paper of Jiang et al. (2018), and the classical biased-sampling sources (Vardi, 1985; Gill et al., 1988). The first gives the additive ordinary-sampling confidence benchmark. Generic spectral-gap concentration from the second incurs the relaxation-time factor avoided by the capped-mass profile. The biased-sampling results are asymptotic and fixed-design.

A live arXiv API, arXiv web, and PMLR search through July 30, 2026 combined pairwise conditional sampling, PCOND, Bradley–Terry–Luce distribution learning, stationary-law estimation, empirical trajectories, total variation, and confidence. It returned the sources above but no theorem with the rate in the main paper, no capped-mass Helmholtz analysis, and no matching high-confidence lower bound. The closest source, Kleinberg et al. (2026), was posted on July 27, 2026 and is cited as the direct starting point rather than presented as concurrent work.

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